

Name: _____

Reg No: _____

Total Marks: 10

1. (3 points) Consider an input feature map of size $7 \times 7 \times 64$ being fed into a convolutional layer with the following specifications:

- Filter size: 3×3
- Number of filters: 128
- Stride: 1
- Padding: "same"

What is the total number of multiplications performed during the forward pass of this layer? (Assume bias addition is negligible for this calculation and show all steps.)

Answer:**Step 1:** Output spatial size with "same" padding and stride 1:

$$\text{Output size} = \left\lceil \frac{7}{1} \right\rceil = 7 \Rightarrow 7 \times 7$$

Step 2: Number of multiplications per output pixel:Each filter has $3 \times 3 \times 64 = 576$ weights. One output pixel = one dot product = 576 multiplications.**Step 3:** Total output pixels: $7 \times 7 \times 128 = 6272$ output values.**Step 4:** Total multiplications: $6272 \times 576 = 3,612,672$ multiplications.

3,612,672

2. (4 points) You are building a simple CNN for handwritten digit recognition. The architecture is as follows:

Layer	Type	Details
1	Input	$32 \times 32 \times 1$ (grayscale)
2	Conv1	6 filters, 5×5 , stride 1, padding 0
3	Pool1	Max pooling, 2×2 , stride 2
4	Conv2	16 filters, 5×5 , stride 1, padding 0
5	Pool2	Max pooling, 2×2 , stride 2
6	Flatten	-
7	FC1	120 neurons
8	FC2	84 neurons
9	Output	10 neurons (softmax)

- (a) (2 points) What is the size of the flattened vector that enters the first fully connected layer? show your working.

Answer:

Step 1: After Conv1: $32 - 5 + 1 = 28$, so $28 \times 28 \times 6$.

Step 2: After Pool1 (2×2 , stride 2): $28/2 = 14$, so $14 \times 14 \times 6$.

Step 3: After Conv2 (5×5 , stride 1, padding 0): $14 - 5 + 1 = 10$, so $10 \times 10 \times 16$.

Step 4: After Pool2 (2×2 , stride 2): $10/2 = 5$, so $5 \times 5 \times 16$.

Step 5: Flattened vector size = $5 \times 5 \times 16 = 400$.

400

- (b) (2 points) Calculate the total number of parameters (weights + biases) in the CONV2 layer only. Show your working.

Answer:

Step 1: Conv2 input depth = 6 (output depth of Conv1/Pool1).

Step 2: Filter size: 5×5 , number of filters = 16.

Weights per filter = $5 \times 5 \times 6 = 150$.

Step 3: Total weights = $150 \times 16 = 2400$.

Step 4: Biases = 1 per filter = 16.

Step 5: Total parameters = $2400 + 16 = 2416$.

2416

3. (3 points) Suppose you have a convolutional layer with 32 filters of size 3×3 operating on an input of size $64 \times 64 \times 16$. If you double the number of filters to 64 but keep everything else the same. How does the number of parameters change? Provide the calculation.

Answer:

Step 1: Parameters in original case:

Weights = $3 \times 3 \times 16 \times 32 = 9 \times 16 \times 32 = 144 \times 32 = 4608$.

Biases = 32.

Total original = $4608 + 32 = 4640$.

Step 2: Parameters after doubling filters to 64:

Weights = $3 \times 3 \times 16 \times 64 = 9 \times 16 \times 64 = 144 \times 64 = 9216$.

Biases = 64.

Total new = $9216 + 64 = 9280$.

Step 3: Change factor: $9280/4640 = 2$.

Number of parameters exactly doubles.

Parameters double from 4640 to 9280